Visual Global-Salient-Guided Network for Remote Sensing Image–Text Retrieval

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***Abstract*— Amid the brisk evolution of remote sensing (RS) technology, the domain of RS cross-modal text–image retrieval (RSCTIR) has captivated scholarly interest for its superior adaptability and symbiotic interaction with human operators. However, due to the heterogeneity between image and text data modalities, feature alignment poses a significant challenge. The existing methodologies overlook the sufficient incorporation of structural guidance during the cross-modal feature interaction alignment process to foster alignment between text and image features. In light of this, we propose an innovative approach for RS image–text retrieval task called visual global-salient-guided network (VGSGN), which comprises two branches: the image branch and the text branch. In the image branch, visual global-salient information sensing module (VGSM) is devised to extract visual global and salient features, aiming to enhance the perception capability for complex backgrounds and scenes in RS images. In the text branch, the textual graph enhancement module (TGEM) is crafted to filter out redundant information in the text features and capture the interactions between words within the text. The design of the multiple visual-guided dynamic fusion (MVGF) module aims to leverage the global and salient features of image to guide the text feature, facilitating cross-modal alignment of text and image features. The experimental results on the widely recognized RSICD and RSITMD datasets corroborate the effectiveness and advancement of our proposed VGSGN in tackling the RSCTIR task.**

***Index Terms*— Remote sensing cross-modal text–image retrieval (RSCTIR), textual graph enhancement, visual-guided dynamic fusion.**

I. Introduction

In RECENT years, researchers pay more and more attention to remote sensing (RS) technology. Massive amounts of RS data contain rich information and have been widely applied in various fields such as land resource acquisition, military reconnaissance, and natural disaster monitoring [1], [2], [3]. How to select valuable knowledge for humans from exponential data is a highly challenging task [4], [5], [6]. Automatic and effective RS image retrieval is receiving unprecedented attention from [7], [8], [9], and [10].

RS cross-modal text–image retrieval (RSCTIR) aims to use text description as query to retrieve RS image. The previous mainstream method of RSCTIR is the caption-based two-stage method [11]. In the first stage, caption is generated for query images, and in the second stage, the similarity between the generated text and the query text is computed to achieve cross-modal image–text retrieval. However, the caption-based two-stage method is essentially a text-to-text retrieval, overlooking the direct matching between RS images and query text. To circumvent the limitations of the two-stage retrieval, in recent years, scholars have proposed the embedding-based single-stage RSCTIR method [12], which directly projects RS image and query text into the same high-dimensional space, enabling the calculation of feature distances and reducing information transformation loss. Therefore, the single-stage retrieval methods that directly align RS images with query text have emerged as the primary methods for RSCTIR.

Currently, the embedding-based RSCTIR methods have shown some progress [13], [14]. However, due to the intrinsic disparities in cross-modal image–text retrieval, significant challenges have been posed in learning image–text embeddings and achieving better alignment of image–text features. There are still several issues that need to be overcome. First, unlike with natural image, RS images typically exhibit characteristics such as uneven distribution of objects, inconsistent target scales, and intricate backgrounds [15], posing a formidable challenge to acquiring visual representation with a robust perception capability. Second, previous methods [16], [17] have focused mainly on extracting text features directly from the entire input query text, neglecting the redundant information inherent within text features and the interactions between words within the query text. Finally, the heterogeneity between modal data of images and text poses challenges in achieving better cross-modal feature alignment. As depicted in Fig. 1, multimodal features inherently possess notable heterogeneous gaps. Existing multimodal feature extraction methods, while to some extent capable of aligning images and text, still exhibit a certain heterogeneous gaps between intra-class images and texts, thereby impacting the results of text–image matching. Existing methods have not provided structural guidance to aid in image–text feature alignment.

In order to overcome these issues, we propose an innovative approach for RS image–text retrieval task called visual global-salient-guided network (VGSGN). VGSGN solves the problem of RSCTIR by focusing on perceiving global and salient information in RS images, redundant filtering and updating enhancement of text features, and leveraging image-guided text to help align image and text features. An illustrative

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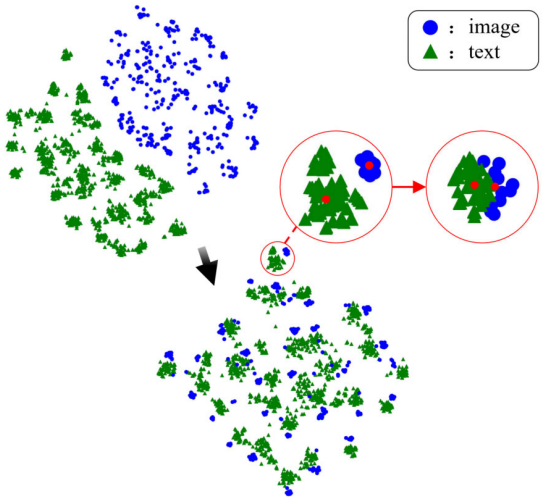


Fig. 1. Upper part represents the multimodal features themselves, and the lower segment represents the multimodal features derived through existing multimodal feature extraction methods. The red encircled region symbolizes the before and after results of aiding in the alignment of text and image features through guided method.

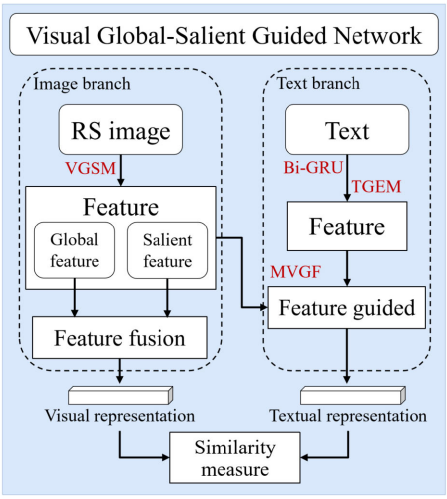


Fig. 2. Illustrative depiction of the VGSGN framework. VGSGN consists of two branches: the image branch and the text branch. The image branch involves feature extraction and feature fusion modules to obtain visual representation. The text branch comprises feature extraction and feature-guided modules to acquire textual representation.

depiction of the VGSGN framework is presented in Fig. 2. This framework primarily comprises two branches: the image branch and the text branch. The image branch comprises a visual global-salient information sensing module (VGSM) and a feature fusion module. The text branch consists of a bidirectional gated recurrent unit (Bi-GRU), a textual graph enhancement module (TGEM), and a multiple visual-guided dynamic fusion (MVGF) module. To enhance the perception capability for complex backgrounds and scenes in RS images, the VGSM is devised to extract visual global and salient features through a multiscale fusion approach. In order to capture the interactions between words within the text more effectively, the TGEM is crafted to filter out redundant information in the text features and update/enhance text features. The design of the MVGF aims to leverage the global and salient features of image to guide the text feature, facilitating cross-modal alignment of text and image features, reducing the distance between image cluster centers and text cluster centers, and achieving better image–text matching results.

The main contributions of our work can be summarized as follows.

1) The proposed MVGF aims to leverage the global and salient features of RS image to guide the text feature during the alignment process between RS image features and text features, facilitating cross-modal alignment of text and image features.
2) The VGSM and TGEM are devised to enhance the ability of image and text branches to extract features. VGSM perceives global and salient information in RS images. TGEM performs redundancy filtering and update enhancement on text features.
3) Our proposed VGSGN has been evaluated on two public datasets, RSICD and RSITMD datasets. The experimental results indicate that the proposed module utilizes image-guided text effectively, and our network achieved state-of-the-art performance.

The subsequent work of this article is organized as follows: in Section II, we briefly introduce the relevant work of RSCTIR, visual–textual semantic alignment, and large models for RS image–text retrieval. Then, in Section III, we provide a detailed introduction to our proposed method and principles. Afterward, we conduct a large number of experiments to verify the effectiveness of the proposed method in Section IV. Finally, the conclusion is given in Section V.

II. Related Work

This section commences by delineating the evolution of RS cross-modal text–image retrieval (RSCTIR), subsequently providing a succinct review of pertinent studies concerning the visual–textual semantic alignment, and finally presents related work on the use of large models for RS image–text retrieval.

A. RS Cross-Modal Text–Image Retrieval

RSCTIR refers to the process of using textual descriptions as queries to retrieve corresponding RS images. Currently, the prevalent approaches to implementing RSCTIR include caption-based methods and embedded-based methods.

The caption-based methods can be categorized as two-stage retrieval methods. Typically, these methods involve the generation of caption text for the queried images through a caption generator. Following the generation of the caption, the similarity between the generated text and the input query text is computed through the BLEU [18]. Qu et al. [19] devised a deep multimodal network for semantic comprehension of high-resolution RS images. Shi and Zou [20] combined deep learning with a fully convolutional network to construct a framework for RS image captioning (RSIC) to better represent RS images. In advancing the RSIC task, Lu et al. [21] introduced comprehensive principles for describing RS images and constructed a large-scale dataset specifically tailored for RSIC tasks. To address overfitting issues within RSIC, Li et al. [22]

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designed a novel truncated cross-entropy loss. A sound-driven attention network [23] is crafted to generate more targeted captions based on the observer's interests. Additionally, Hoxha and Melgani [24] designed a support vector machine-based decoder to address the high computational demands of current caption generators when training data is limited. Although caption-based RSCTIR methods have developed relatively maturely, the inherent two-stage constraint inevitably leads to information loss, consequently impacting retrieval accuracy.

The embedded-based methods involve directly mapping the data from the two modalities, images and text, onto the same high-dimensional space to measure the similarity between different modality features. Abdullah et al. [25] designed an RSCTIR framework based on a deep bidirectional triplet network to learn the joint encoding of multimodal data, aiming to acquire more robust embedding. Addressing the multiscale and target redundancy issues in RS images, Yuan et al. [13] introduced an asymmetric multiscale feature matching network and provided a fine-grained RS image–text dataset. Acknowledging the significance of local information, Yuan et al. [14] designed a dynamic fusion module to merge global and local features within the images. To alleviate the issue of information imbalance between different modalities, Zhang et al. [26] proposed the KEA mechanism and introduced CL learning to promote feature-adaptive alignment. Pan et al. [27] designed a fine-grained perception module to enhance scene perception abilities and reduce semantic confusion. Despite the progress made by the embedded-based RSCTIR methods, previous approaches have overlooked the importance of salient features in RS image and the influence of individual words on others within text.

B. Visual–Textual Semantic Alignment

In the sphere of image–text retrieval tasks, owing to the inherent heterogeneity of image and text data modalities, the representations of data across different modalities are distinct and occupy separate feature spaces. A central issue presents itself as how to achieve semantic alignment between image and text features within a shared embedding space. Park et al. [28] devised a cross-modal semantic alignment module that enhances the relationships between modalities to establish an explicit alignment of features. Lee et al. [29] proposed stacked cross-attention (CA) to detect latent alignments between image regions and words within the text, thus inferring image–text similarity. Chen et al. [30] constructed two separate embedding spaces and introduced semantic consistency constraints to foster feature alignment. Cheng et al. [31] designed a semantic alignment module to probe into the potential correspondences between RS image and text, securing feature representations with enhanced discriminative capabilities. Liu et al. [32] introduced a graph-structured matching network to form graph structures of images and text, employing graph convolution to propagate nodes, which facilitates the inference of granular phrase correspondences. Wang et al. [33] presented a fusion-layer-based method for interactivity between image and text features to extract the relationship between cross-modal features. Yao et al. [34] proposed an image-induced multimodal fusion module, designed to establish feature associations between RS images and query text. However, throughout the course of cross-modal feature interaction alignment, previous methods did not use image features structurally to guide text feature, in order to facilitate the alignment between image and text features.

C. Large Models for RS Image–Text Retrieval

Radford et al. [35] proposed the CLIP model, which has demonstrated strong multimodal image–text retrieval capabilities in the natural image domain. Using vision-language large models such as CLIP to address the RS image–text retrieval problem also shows promising prospects. Large model-based methods can be broadly categorized into two principal avenues. One is using large-scale datasets to retrain the large models. Wang et al. [36] constructed a large and semantically diverse vision-language dataset (SkyScript) and performed continual pre-training on SkyScript to obtain a remote-sensing-specialized CLIP model, denoted as SkyCLIP. Liu et al. [37] constructed a large-scale dataset (including RSICD and RSITMD) to retrain CLIP, which achieves excellent performance. However, the computational demands incurred by retraining large models are substantial. The other is fine-tuning of large models. Currently fine-tuning large model has become a research hotpot. The quality of the fine-tuning method will have a direct impact on the performance of the fine-tuned large model. Achieving highly effective fine-tuning is fraught with challenges and requires in-depth research.

III. Method

This section will introduce our proposed method VGSGN, and we show its overall structure in Fig. 3. We mainly introduce our method from four aspects: formulation in Section III-A, feature enhancement (VGSM, TGEM) in Section III-B, and the MVGF in Section III-C. Finally, we present the objective function and training procedure in Section III-D.

A. Formulation

For RSCTIR, given an input RS image I and an input sentence T. The model extracts image features and text features, respectively, and maps them to the same space for similarity calculation. This process can be represented as

$$\mathcal{V} = W_v I, \quad \mathcal{T} = W_t T$$
$$S = \cos(\mathcal{V}, \mathcal{T}) \tag{1}$$

where W_v and W_t are the weight matrix of the visual and textual representations, respectively, and $\mathcal{V}$ and $\mathcal{T}$ represent the visual and textual representations, respectively. $\cos(x, y)$ denotes the cosine similarity of vectors x and y. We will introduce the overall flow of our method from the visual representation and textual representation.

1) Visual Representation: We use traditional CNN for extracting multiscale visual features from RS images. Consistent with the approach in [27], we use ResNet pre-trained on the AID dataset [38] as an image feature extractor and transfer its parameters to our task. The visual

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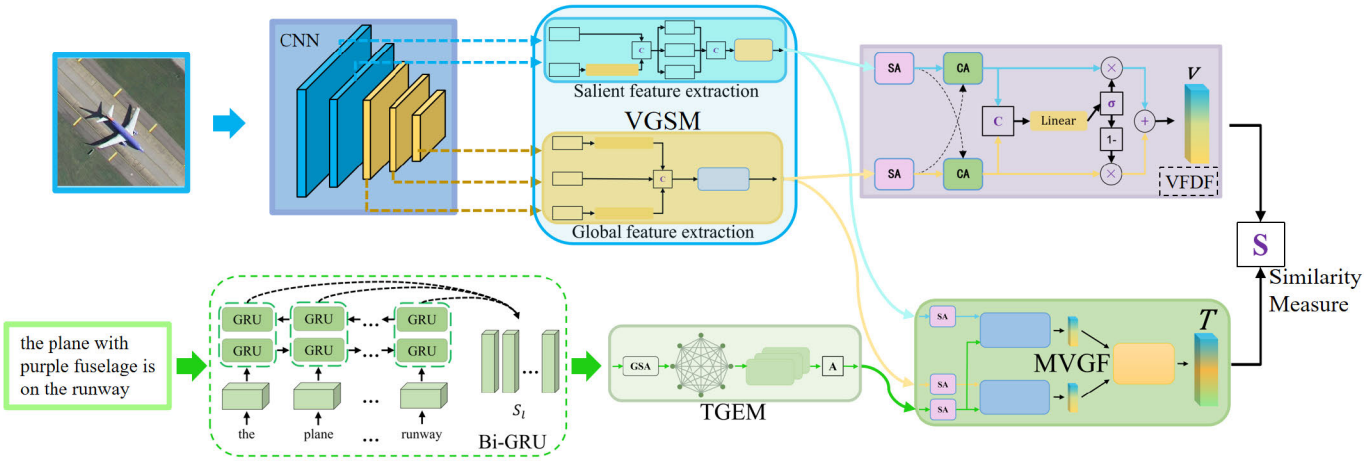


Fig. 3. Overall structure of our proposed VGSGN. In the visual representation portion, we first use traditional CNN to extract the output features of each layer, then extract global and salient features of the image through the VGSM, and finally fuse the features through the VFDF module to obtain the final visual representation. In the text representation portion, Bi-GRU is first used to extract the primitive text feature, then the TGEM is used to enhance the text feature, and finally visual features are used in the MVGF module to guide the text feature to obtain the final textual representation.

features of the xth layer obtained using the image feature extractor are represented as $V^x (x = 1, 2, 3, 4, 5)$.

Considering that RS images usually exhibit characteristics such as uneven distribution of objects, inconsistent target scales, and intricate backgrounds, VGSM is designed to extract visual global and salient features through a multiscale fusion approach to enhance the network's perception capability for complex backgrounds and scenes in RS images

$$V_g, V_s = \text{VGSM}(V^x) \tag{2}$$

where V_g and V_s are the visual global and salient features of the VGSM obtained by the multiscale fusion method. The details of VGSM are presented in Section III-B1. Then we fuse the visual global and salient features to get the final visual representation $\mathcal{V}$

$$\mathcal{V} = \text{VFDF}(V_g, V_s) \tag{3}$$

where VFDF denotes the visual feature dynamic fusion, and the relevant details are in Section III-B2.

2) Textual Representation: For a sentence T containing L words, denoted $\{w_i\}_{i=1}^{L}$, w_i represents the ith word. We first map each word to the corresponding word vector [39] via an embedding matrix

$$e_i = W_e(w_i) \tag{4}$$

where W_e represents the word embedding layer, and e_i represents the embedding vector of the ith word.

Similar to the previous method [13], [14], [26], considering the contextual and temporal information in the sentence, we use Bi-GRU [40] to extract semantic features. Each word is sequentially input into the bidirectional GRU in different steps. The formulae are expressed as follows:

$$\begin{aligned} \overrightarrow{h_i} &= \overrightarrow{\text{GRU}}(e_i, \overrightarrow{h_{i-1}}) \\ \overleftarrow{h_i} &= \overleftarrow{\text{GRU}}(e_i, \overleftarrow{h_{i+1}}) \\ h_i &= \frac{\overrightarrow{h_i} + \overleftarrow{h_i}}{2} \end{aligned} \tag{5}$$

where $\overrightarrow{\text{GRU}}(\cdot)$ and $\overleftarrow{\text{GRU}}(\cdot)$ represent the forward and the backward GRU. $\overrightarrow{h_i}$ and $\overleftarrow{h_i}$ indicate the state of the $\overrightarrow{\text{GRU}}$ and $\overleftarrow{\text{GRU}}$ at step i. h_i represent the textual representation at the ith step. From these, we can obtain the initial text feature $S_l = [h_1, h_2, \ldots, h_L]^T$.

In order to better capture the interactions between words within the text, TGEM is designed to filter redundant information in text features as well as to update and enhance text features using graph networks

$$T_e = \text{TGEM}(S_l) \tag{6}$$

where T_e represents the text features obtained by TGEM enhanced by redundant filtering as well as graph network updating. The details of TGEM are presented in Section III-B3. Finally, MVGF is proposed to obtain the final text representation $\mathcal{T}$ using visual global and salient features to guide the text features

$$\mathcal{T} = \text{MVGF}(V_g, V_s, T_e). \tag{7}$$

The details of MVGF are introduced in Section III-C.

B. Feature Enhancement

1) Visual Global-Salient Information Sensing Module (VGSM): Traditional cross-modal retrieval methods [41], [42] generally only apply a fully connected layer to the output features of the last layer to map the visual features to the target-sized feature vectors, but, as the network deepens, the information related to the small targets in the extracted features will gradually decrease or even disappear. To better represent visual features, VGSM is designed to extract visual global and salient features through a multiscale fusion approach. We believe that deeper features in the network contain more visual global information and shallow features contain more visual salient spatial information. As shown in Fig. 4, we use three 1×1 convolution layers to process the different deep features as

$$\hat{V}^x = \text{Conv}_{2d}^x(V^x) \tag{8}$$

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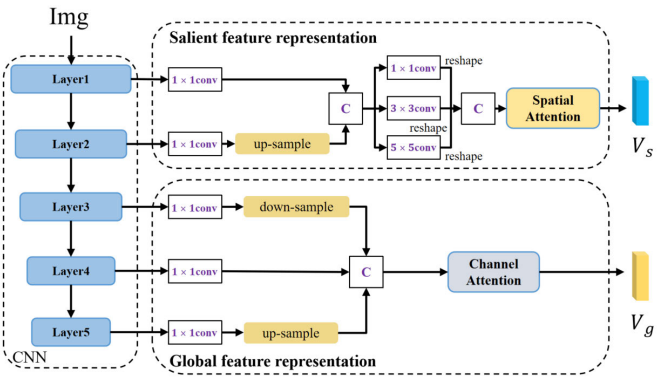


Fig. 4. VGSM. Salient feature representation: connect the first and second layers of CNN outputs, use fine-grained perception and spatial attention modules to extract image salient feature. Global feature representation: connect the third, fourth, and fifth layers of CNN outputs, and use the channel attention module to extract image global feature.

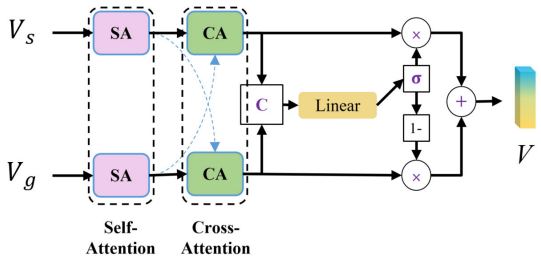


Fig. 5. VFDF. Use SA and CA modules to interact the salient feature and global feature of the RS image, and then obtain the final visual feature representation through a designed dynamic fusion method.

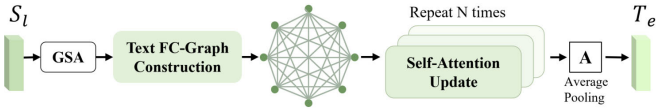


Fig. 6. TGEM. Initially, the GSA module is employed to filter out redundant information from the input text features, followed by constructing a fully connected text graph to update and enhance the text features through an SA mechanism.

where $\text{Conv}_{2d}^{x}(\cdot)$ denotes 2-D convolution for the xth deep feature. For global feature, we sample the last three features of different sizes and connect them as

$$V_d = \text{Concat}(\text{Downsample}(\hat{V}^3), \hat{V}^4, \text{Upsample}(\hat{V}^5)) \quad (9)$$

where V_d represents deep multiscale visual features. Downsample$(\cdot)$ and Upsample$(\cdot)$ denote down-sampling and up-sampling operations. Then, to better grasp the global information, we perform channel attention [43] on V_d to obtain the global visual feature V_g.

For salient feature, similarly, we sample the first two features of different sizes and connect them as

$$V_l = \text{Concat}(\hat{V}^1, \text{Upsample}(\hat{V}^2)) \quad (10)$$

where V_l represents shallow multiscale visual features that contain more visual salient spatial information. To enhance the perception of various fine-grained spatial features, we employ convolution kernels of different sizes $(1 \times 1, 3 \times 3, 5 \times 5)$ to extract V_l as

$$\mathcal{F}_e^j = \text{Reshape}(\sigma(\text{BN}(\text{Conv}_{2d}^{j}(V_l)))) \quad (11)$$

where $\mathcal{F}_e^j$ represents the extracted feature, Reshape$(\cdot)$ represents reshaping (H, W) 2-D into 1-D, σ denotes the sigmoid function, and $\text{Conv}_{2d}^{j}(\cdot)$ represents the jth convolution kernel with different sizes $(j = 1, 2, 3)$. Then, we directly integrate extracted features through connection as

$$V_e = \text{Concat}(\mathcal{F}_e^1, \mathcal{F}_e^2, \mathcal{F}_e^3) \quad (12)$$

where V_e represents the multigranularity perception feature. To refine the network's focus on processing spatial information, we incorporate spatial attention [43] on V_e to obtain the global visual feature V_s.

2) Visual Feature Dynamic Fusion: To promote interaction between global feature and salient feature for a better visual representation, as shown in Fig. 5, we first use self-attention (SA) and cross-attention (CA) to process features as

$$\begin{aligned} V_g^{\text{SA}} &= \text{SA}(V_g),\ V_s^{\text{SA}} = \text{SA}(V_s) \\ V_g^{\text{CA}} &= \text{CA}(V_g^{\text{SA}}, V_s^{\text{SA}}),\ V_s^{\text{CA}} = \text{CA}(V_s^{\text{SA}}, V_g^{\text{SA}}) \end{aligned} \quad (13)$$

where $\text{SA}(x)$ represents using SA to process the feature x. $\text{CA}(x, y)$ represents using x as the query and y as the key and value for interaction in attention mechanism. V_g^{SA} and V_s^{SA} represent the global and salient features processed by the SA module. V_g^{CA} and V_s^{CA} represent the global and salient features after feature interaction. Then, we fuse global feature with salient feature to obtain the comprehensive visual feature. We concatenate V_g^{CA} and V_s^{CA} and perform it with a linear neural network to obtain the learnable weight vector v_w for generating the fusion feature $\mathcal{V}$

$$\begin{aligned} v_w &= \text{Linear}(\text{Concat}(V_g^{\text{CA}}, V_s^{\text{CA}})) \\ \mathcal{V} &= (V_s^{\text{CA}} \odot \sigma(v_w)) \oplus (V_g^{\text{CA}} \odot (1 - \sigma(v_w))) \end{aligned} \quad (14)$$

where v_w represents the weighted vector, Linear$(\cdot)$ represents a linear neural network, and $\mathcal{V}$ represents the final visual feature representation of the image.

3) Textual Graph Enhancement Module: The previous approaches [13], [14] have focused on extracting text features directly from the entire input query text, neglecting the redundant information inherent within text features and the interactions between words within the query text. In order to better capture the interactions between words within the text, TGEM is designed to filter redundant information in text features as well as to update and enhance text features using graph networks. As shown in Fig. 6, similar to [44], we use gating SA (GSA) to filter redundant information in text features. The structure of the GSA module is shown in Fig. 7.

Specifically, setting the input of the GSA module as X, the GSA module can be expressed as

$$\text{GSA}(X) = \text{Concat}(h_1, \ldots, h_H) \quad (15)$$

where $h_i =$

$$\text{Softmax}\left(\frac{(M_i^Q \odot Q_i)(M_i^K \odot K_i)^T}{\sqrt{d}}\right) V_i \quad (16)$$

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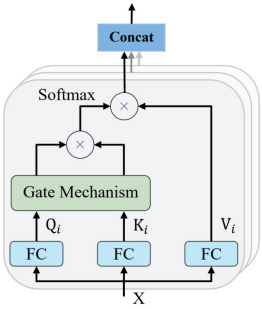


Fig. 7. Gating SA. Within the SA mechanism, a gating function is introduced to focus more keenly on important information, thereby diminishing the intrusion of unnecessary data.

where $Q_i = XW_i^Q$, $K_i = XW_i^K$, and $V_i = XW_i^V$ are the query, key, and value. W_i^Q,W_i^K, and W_i^V are the corresponding learnable projection matrix. Softmax$(\cdot)$ represents using softmax function to activate the input feature. $\odot$ represents the Hadamard product of matrix. M_i^Q and M_i^K represent the gating masks in the gating mechanism, and their calculation process is as follows:

$$\begin{aligned} M_i^Q &= \sigma(G_i W_M^Q + b_M^Q) \\ M_i^k &= \sigma(G_i W_M^K + b_M^K) \\ G_i &= (Q_i W_G^Q + b_G^Q) \odot (K_i W_G^K + b_G^K) \end{aligned} \tag{17}$$

where W_M^Q,W_M^K,b_M^Q, and b_M^M are the learnable parameters of fully connected layers. $W_M^Q(W_G^K)$ and $b_G^Q(b_G^K)$ represent the learnable projection matrices and bias vectors.

After GSA module processing, the feature of filtered redundant information is as follows:

$$S_l^G = \mathrm{GSA}(S_l) = \left[h_1^G, h_2^G, \ldots, h_L^G\right]^T. \tag{18}$$

Then use S_l^G as the vertices of the graph and connect any paired vertices with an edge to construct the textual fully connected graph with L vertices. The contribution between words determines the weight of edges, and the calculation process is as follows:

$$\begin{aligned} \omega_i &= W_i^T t_i + W_v^T h + b_i \\ \hat{\omega}_j &= \exp(\omega_j)/\textstyle\sum_{i=1}^{L}\exp(\omega_i) \end{aligned} \tag{19}$$

where $h \in S_l^G$ is a vertex in the textual fully connected graph, $t_i \in S_l^G$ is the ith vertex connected to h, and $\hat{\omega}_j$ represents the weight of the edges connected to h and t_i.

After calculating the weights of each edge in the textual fully connected graph, each vertex (word) can update itself based on the importance of other vertices (words)

$$\hat{h_i^G} = \sum_{j=1}^{L} \hat{\omega}_j t_j \tag{20}$$

where $\hat{h_i^G}$ is the ith vertex feature after update. In order to enable words to learn more information from other words in query text, we repeat the above process N times to obtain enhanced text features $\hat{S_l^G} = \left[\hat{h_1^G}, \hat{h_2^G}, \ldots, \hat{h_L^G}\right]^T$. Then perform average processing on $\hat{S_l^G}$ to obtain enhanced sentence feature representation T_e

$$T_e = \frac{1}{L}\sum_{i=1}^{L} \hat{h_i^G}. \tag{21}$$

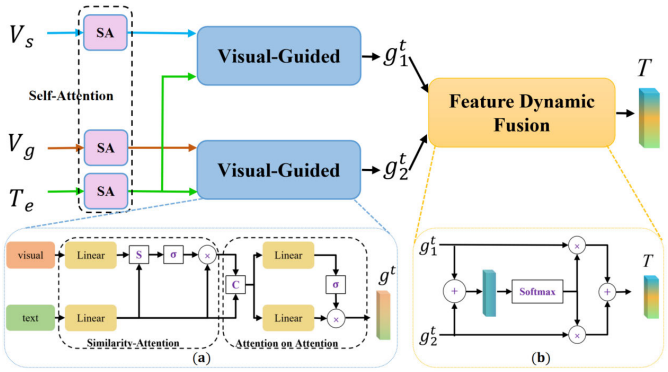


Fig. 8. MVGF module. First, the SA module processes the visual salient feature, the visual global feature, and the text feature. Thereafter, the visual salient feature and global feature individually guide the text feature through the VGM. Finally, the final text representation is secured through a meticulously designed dynamic fusion module. (a) VGM. (b) Feature dynamic fusion module.

C. MVGF Module

The heterogeneity between the two modal data, that is, image and text, makes cross-modal feature alignment difficult. Existing methods [14], [27], where the image and text branches are independent, are not conducive to achieving image–text feature interaction. The design of the MVGF aims to leverage image-guided text to structurally establish connections between image and text features, thereby facilitating cross-modal alignment of image and text features, reducing the distance between image cluster centers and text cluster centers, and achieving better text–image matching results. The specific structure of MVGF is shown in Fig. 8. We first use the SA module to process the input features, then guide the text features using visual global features and salient features, respectively, and finally dynamically fuse the features to obtain the final text feature representation. We will provide a detailed introduction to the specific modeling process.

1) Visual-Guided Module (VGM): Inspired by [45] and [46], we design VGM with the structure shown in Fig. 8(a), aiming to use visual feature v to guide text feature t. It can be expressed as follows:

$$g^t = \mathrm{VGM}(v, t). \tag{22}$$

Specifically, first perform similarity attention on visual and text feature

$$t^{vg} = \sigma(S(\tilde{v}, \tilde{t})) \odot \tilde{t} \tag{23}$$

where $\tilde{v}$ and $\tilde{t}$ are obtained from v and t through linear layer, and $S(x, y)$ represents the similarity between x and y. t^{vg} represents the features processed by similarity attention. Then, combining t^{vg} and $\tilde{t}$ to generate denoising SA with the gate mechanism

$$g^t = \mathrm{linear}(\mathrm{Concat}(t^{vg}, \tilde{t})) \odot \sigma(\mathrm{linear}(\mathrm{Concat}(t^{vg}, \tilde{t}))) \tag{24}$$

where g^t represents the visual-guided textual feature.

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Therefore, we can obtain the visual-guided textual features with visual global feature and visual salient feature

$$g_1^t = \mathrm{VGM}(V_s, T_e)$$
$$g_2^t = \mathrm{VGM}(V_g, T_e). \tag{25}$$

2) Textual Feature Dynamic Fusion: Another function of the MVGF module is the dynamic fusion of multiple visual-guided textual features. As shown in Fig. 8(b). In order to generate dynamic weights based on different features, we first superimpose these features to obtain the feature t_{mix}, and then perform a linear transformation on t_{mix} to obtain the learnable dynamic weights λ, which are used to generate the fused feature $\mathcal{T}$

$$t_{\mathrm{mix}} = g_1^t \oplus g_2^t$$
$$\lambda_1, \lambda_2 = \mathrm{Softmax}(\sigma(t_{\mathrm{mix}} W_\alpha) W_\beta)$$
$$\mathcal{T} = (\lambda_1 \otimes g_1^t) \oplus (\lambda_2 \otimes g_2^t) \tag{26}$$

where W_α and W_β are the learnable weight matrices. $\mathcal{T}$ represents the final textual feature representation of the query text.

D. Objective Function and Training Procedure

1) Objective Function: In the field of cross-modal RS retrieval, the triplet loss, as one of the mainstream loss functions for multimodal feature matching, is often used for feature alignment. The triplet loss aims to make the distance between the sample and the positive sample as close as possible, while also making the distance between the sample and the negative sample as far as possible. Following [13], [29], it can be modeled as

$$\mathcal{L}_{(\mathcal{V},\mathcal{T})} = \sum_{\widehat{\mathcal{T}}} [m - S(\mathcal{V}, \mathcal{T}) + S(\mathcal{V}, \widehat{\mathcal{T}})]_+ + \sum_{\widehat{\mathcal{V}}} [m - S(\mathcal{V}, \mathcal{T}) + S(\widehat{\mathcal{V}}, \mathcal{T})]_+ \tag{27}$$

where m represents the margin parameter, $[x]_+ \equiv \max(x, 0)$ and $S(\mathcal{V}, \mathcal{T})$ represent the similarity of the paired RS image feature $\mathcal{V}$ and text feature $\mathcal{T}$. $\widehat{\mathcal{T}}$ is the text feature in a single batch that is not paired with RS image feature $\mathcal{V}$, and $\widehat{\mathcal{V}}$ is the RS image feature in a single batch that is not paired with text feature $\mathcal{T}$. Throughout the training process, we use the triplet loss $\mathcal{L}_{(\mathcal{V},\mathcal{T})}$ as the objective function.

2) Training Procedure: The training procedure of the proposed VGSGN is shown in Algorithm 1. During the training process, the paired RS image–text pairs P_{it} are needed. For the RS image feature representation, we first extract RS image features using pre-trained ResNet-50 [47]. Then, VGSM is used to obtain the global feature V_g and salient feature V_s of the RS image, and the VFDF module fuses the global feature V_g and salient feature V_s to obtain the visual feature representation $\mathcal{V}$. For the text feature representation, Bi-GRU is first used to extract feature S_l, followed by TGEM for redundant information filtering and self-attention updating to obtain enhanced text feature T_e. Finally, MVGF guides T_e using the global feature V_g and salient feature V_s of RS image to obtain the text feature representation $\mathcal{T}$. The triplet loss function calculates the loss l incurred from the distance between visual feature $\mathcal{V}$ and text feature $\mathcal{T}$, and utilizes l to optimize the various submodules of VGSGN.

Algorithm 1 VGSGN Training Procedure

Require:
RS training dataset $P_{\mathrm{it}} = \{(i_1, t_1), (i_2, t_2), \ldots\}$ (i_n, t_n) is the paired RS image–text pair
// Image section
CNN with VGSM $V_s, V_g = \mathcal{C}_{VGSM}(i; \theta_1)$
VFDF module $\mathcal{V} = \mathcal{F}_{VFDF}(V_s, V_g; \theta_2)$
// Text section
GRU module $S_l = \text{Bi-GRU}(t; \theta_3)$
TGEM module $T_e = \mathrm{TGEM}(S_l; \theta_4)$
MVGF module $\mathcal{T} = \mathrm{MVGF}(V_s, V_g, T_e; \theta_5)$
Repeat until convergence:

for each batch $i, t \in P_{\mathrm{it}}$ **do**
  // Visual feature representation
  $V_s, V_g = \mathcal{C}_{VGSM}(i; \theta_1)$
  $\mathcal{V} = \mathcal{F}_{VFDF}(V_s, V_g; \theta_2)$
  // Textual feature representation
  $S_l = \text{Bi-GRU}(t; \theta_3)$
  $T_e = \mathrm{TGEM}(S_l; \theta_4)$
  $\mathcal{T} = \mathrm{MVGF}(V_s, V_g, T_e; \theta_5)$
  // Calculate the triplet loss
  $l = \mathcal{L}_{(\mathcal{V},\mathcal{T})}$
  Update θ_1, θ_2, θ_3, θ_4, θ_5 **by** l
end for
return $\mathcal{C}_{VGSM}$, $\mathcal{F}_{VFDF}$, Bi-GRU, TGEM, MVGF

IV. Experiments Results and Analysis

In this section, we first introduce the datasets employed and the evaluation metrics. We then detail some experimental nuances. Following this, we engage in comparisons and analyses with various RSCTIR methods across several RS image–text datasets to affirm the effectiveness of the proposed VGSGN. In addition, we conduct a comprehensive suite of ablation experiments to systematically examine the influence of each submodule within VGSGN. Feature visualization experiments are also performed to intuitively analyze the role of the MVGF module in facilitating cross-modal semantic alignment. Then, we provide comparative studies on image–text retrieval tasks and semantic localization tasks. We also compare the retrieval performance with CLIP-based methods. Finally, time consumption is compared.

A. Dataset and Evaluation Metrics

In the experiment, we use two common RS image–text datasets in RSCTIR: RSICD [21] and RSITMD [13]. As shown in Fig. 9, each RS image–text sample in these two datasets consists of one RS image and five corresponding text description sentences. The RSICD dataset contains 10 921 pairs of samples, and the size of RS images is 224×224. The RSITMD dataset has fewer samples than RSICD, containing 4743 pairs of samples. The size of RS images is 256×256, but, in terms of text, it has a more fine-grained representation

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- there is a airport in the middle and many planes in it .
- a silver building in the middle while with many plane besides .
- many black road on the either hand of the airport .
- many dark buildings besides the silver building while with many plane besides .
- many planes are parked near a large building in an airport .

(a)



- On one side of a deep black river is a blue factory .
- Blue boats, white boats and green boats are moored by the black river.
- There were many cars parked on the bank of the river where the three boats were moored.
- Many cars were parked on the bank of the river where three boats were moored.
- Blue boats, white boats and green boats are anchored on one side of the black river respectively.

(b)

Fig. 9. Sample examples of RSICD and RSITMD datasets. Each sample consists of one RS image and five corresponding text description sentences. (a) RSICD sample. (b) RSITMD sample.

than RSICD. Although there are two common datasets: UCM Captions and Sydney, we do not use them due to their small sample size of only 2000 and 600, respectively. In the experiment, we divide the dataset into training set, validation set, and test set in a ratio of 8:1:1, which is similar to [13].

Similar to previous cross-modal retrieval methods [13], [14], [26], [27], we use R@k and mR [48] evaluation metrics to evaluate the performance of the model. R@k is the ratio of the ground truth in the top k recall samples, with k taken as 1, 5, and 10 to comprehensively evaluate the results. mR is the average value of multiple R@k, which can more intuitively reflect the overall performance of the model.

B. Implementation Details

All experiments in this article are implemented by the Python programming language and PyTorch [49] framework, conducted on an NVIDIA RTX 3090 GPU. For the sake of equitable comparison, our method adopts the same type and conventional data augmentation as previous methods. The data pre-processing and feature space dimensionality are consistent with [14], [26]. For data pre-processing, we uniformly scale the image size from different datasets to 256 × 256 in the experiment, and followed by employing rotation and flipping as means of data augmentation to improve the robustness of the model. For the feature space dimensionality, the dimension of a single word vector is set to 300, and the dimensions of image feature vectors and text feature vectors are 512. The margin loss of the triplet is set to 0.2, the number of the parallel processing layers of the SA and CA modules is set to 4, the number of the parallel processing layers of the GSA module is set to 2, and the repetition number N of the text graph network enhancement module is set to 4. We use Adam [50] as the model optimizer. We totally train 70 epochs and set the batch size to 128. The initial learning rate is set to 0.0002 and decays by 0.7 every 20 periods. In addition, to improve the persuasiveness of the experiment, same as [26], we conduct three consecutive training and verification for each group of experiments to take the average result.

C. Comparisons With Several SOTA Methods

We have compared the proposed VGSGN with several previous advanced RSCTIR methods on the RSICD and RSITMD datasets. These methods can be classified into two categories: traditional cross-modal retrieval methods, including VSE++, SCAN, CAMP, and MTFN, and RS cross-modal retrieval methods, such as LW-MCR, AMFMN, GaLR, HVSA, and SWAN. The results are as illustrated in Table I; the portion above the dashed line represents conventional cross-modal retrieval approaches, whereas the segment below pertains to RS cross-modal retrieval methods. Within each criterion, the bolded figures denote the most favorable outcomes, while those underlined signify the second-best performances.

1) *VSE++ [51]:* VSE++ focuses on challenging instances, leveraging CNN and GRU for feature extraction, and computing the distance between features using a triplet loss function.
2) *SCAN [29]:* SCAN introduces a cross-modal attention mechanism between image regions and words to align targets in the images with those in the text.
3) *CAMP [52]:* CAMP has devised an information dissemination approach that adaptively controls the transmission of model information, employing an attention module for feature alignment.
4) *MTFN [53]:* MTFN leverages the fusion of multimodal characteristics, employing low-rank decomposition to compute distances between features.
5) *LW-MCR [54]:* A lightweight retrieval model utilizing knowledge distillation and contrastive learning.
6) *AMFMN [13]:* AMFMN employs the guidance of image features to fuse sentence and keyword features, thereby enhancing textual feature representation.
7) *GaLR [14]:* GaLR utilizes a CNN to extract global features and an object detector to extract local features, integrating both to obtain the image feature representation.
8) *HVSA [26]:* HVSA has devised the KEA mechanism to alleviate information imbalances between different modalities and applies CL learning to achieve adaptive alignment of features.
9) *SWAN [27]:* SWAN has developed a fine-grained perception module to enhance scene awareness in order to reduce semantic confusion.

The comprehensive results clearly demonstrate that the overall performance of traditional cross-modal retrieval methods is inferior to RS cross-modal retrieval methods, indicating the impracticality of directly applying traditional cross-modal retrieval methods to the realm of RS. Across both the RSICD and RSITMD datasets, our approach has consistently achieved superior outcomes. Specifically, on the RSICD dataset, the VGSGN method garnered an mR rating of 21.55, surpassing the leading competitor by 0.94. Additionally, on the RSITMD dataset, the VGSGN method demonstrated even more remarkable performance with an mR score of 36.50, surpassing the leading competitor by 2.39. This could be attributed to the RSITMD dataset having a more fine-grained representation compared to the RSICD dataset. Through the TGEM and MVGF modules, VGSGN is able to obtain superior textual feature representations, thereby enhancing the overall performance of cross-modal retrieval. In practical application, as the volume of data expands, the presence of the VGM incurs

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TABLE I
COMPARISONS OF RETRIEVAL PERFORMANCE ON RSICD AND RSITMD TEST SET

Approach	RSICD dataset							RSITMD dataset						
	Sentence Retrieval			Image Retrieval			mR	Sentence Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
VSE++ [51]	3.38	9.51	17.46	2.82	11.32	18.10	10.43	10.38	27.65	39.60	7.79	24.87	38.67	24.83
SCAN t2i [29]	4.39	10.90	17.64	3.91	16.20	26.49	13.25	10.18	28.53	38.49	10.10	28.98	43.53	26.64
SCAN i2t [29]	5.85	12.89	19.84	3.71	16.40	26.73	14.23	11.06	25.88	39.38	9.82	29.38	42.12	26.28
CAMP-triplet [52]	5.12	12.89	21.12	4.15	15.23	27.81	14.39	11.73	26.99	38.05	8.27	27.79	44.34	26.20
CAMP-bce [52]	4.20	10.24	15.45	2.72	12.76	22.89	11.38	9.07	23.01	33.19	5.22	23.32	38.36	22.03
MTFN [53]	5.02	12.52	19.74	4.90	17.17	29.49	14.81	10.40	27.62	36.28	9.96	31.37	45.84	26.92
LW-MCR(b) [54]	4.57	13.71	20.11	4.02	16.47	28.23	14.52	9.07	22.79	38.05	6.11	27.74	49.56	25.55
LW-MCR(d) [54]	3.29	12.52	19.93	4.66	17.51	30.02	14.66	10.18	28.98	39.82	7.79	30.18	49.78	27.79
AMFMN-soft [13]	5.05	14.53	21.57	5.05	19.74	31.04	16.02	11.06	25.88	39.82	9.82	33.94	51.90	28.74
AMFMN-fusion [13]	5.39	15.08	23.40	4.90	18.28	31.44	16.42	11.06	29.20	38.72	9.96	34.03	52.96	29.32
AMFMN-sim [13]	5.21	14.72	21.57	4.08	17.00	30.60	15.53	10.63	24.78	41.81	11.51	34.69	54.87	29.72
GaLR [14]	6.59	19.85	31.04	4.69	19.48	32.13	18.96	**14.82**	31.64	42.48	11.15	36.68	51.68	31.41
HVSA [26]	7.47	20.62	32.11	5.51	21.13	34.13	20.16	13.20	32.08	45.58	11.43	39.20	57.45	33.16
SWAN [27]	7.41	20.13	30.86	5.56	22.26	**37.41**	20.61	13.35	32.15	46.90	11.24	40.40	60.60	34.11
VGSGN	**8.33**	**21.87**	**32.57**	**6.53**	**23.13**	36.85	**21.55**	14.16	**34.96**	**50.66**	**13.23**	**42.57**	**63.41**	**36.50**

TABLE II
ABLATION EXPERIMENT FOR DIFFERENT STRUCTURES OF VGSGN ON RSITMD TEST SET

Abalation Model	VGSM		TGEM	MVGF		Sentence Retrieval			Image Retrieval			mR
	V_s	V_g	T_e	g_1^t	g_2^t	R@1	R@5	R@10	R@1	R@5	R@10	
$a1$	√		√			5.79	19.25	30.09	5.66	24.69	40.49	21.03
$a2$	√		√	√		9.96	20.35	30.10	7.04	26.06	41.02	22.02
$a3$		√	√			11.44	31.19	45.45	11.21	36.96	54.80	31.84
$a4$		√	√		√	11.72	32.41	45.58	12.41	38.43	56.74	32.88
$a5$			√			10.39	25.87	39.25	9.96	34.52	50.43	28.40
$a6$	√	√		√	√	11.95	31.64	47.57	12.61	42.57	62.52	34.81
$a7$	√	√	√			12.83	30.09	44.03	10.31	37.70	57.08	32.01
full	√	√	√	√	√	**14.16**	**34.96**	**50.66**	**13.23**	**42.57**	**63.41**	**36.50**

greater resource consumption during the re-establishment of the correspondences between images and texts. Consequently, our methodology is primarily tailored for scenarios where the data amounts are not excessively large, yet the requirements for accuracy are particularly stringent.

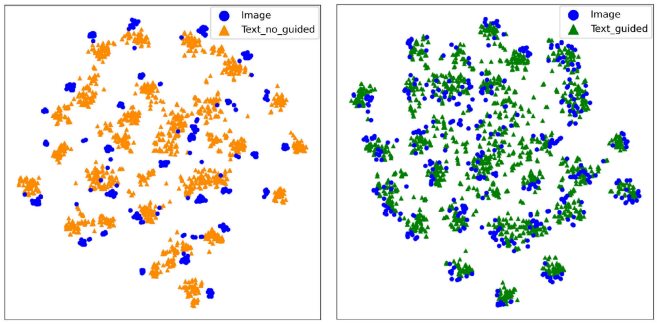
D. Ablation Studies

In this section, a series of experiments are conducted to systematically analyze the proposed VGSGN. We design seven ablation experiments on the RSITMD dataset to investigate the impact of each module on retrieval performance, as shown in Table II.

1) $a1$: Solely utilizing visual salient feature as the image feature representation, without guiding text features through the single VGM.
2) $a2$: Solely utilizing visual salient feature as the image feature representation, guiding text features through the single VGM.
3) $a3$: Solely utilizing visual global feature as the image feature representation, without guiding text features through the single VGM.
4) $a4$: Solely utilizing visual global feature as the image feature representation, guiding text features through the single VGM.
5) $a5$: Utilizing the output of the last layer of CNN as image feature representation, without guiding text feature through MVGF module.
6) $a6$: Not employing the TGEM for redundancy filtering and attention updating of text features.
7) $a7$: Fusing visual global and salient feature to obtain the image feature representation, without guiding text feature through MVGF module.
8) $Full$: Complete VGSGN structure.

In these models, the comparison of results between $a1$ and $a2$, as well as between $a3$ and $a4$, reveals the impact of single VGM on the experimental outcomes. The comparison between $a1$ and $a3$, as well as between $a2$ and $a4$, indicates that

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(a) (b)

Fig. 10. Feature visualization. We employ the t-SNE algorithm to embed image and text feature onto a 2-D plane from the latent feature space. The outcomes are illustrated as follows: blue denotes image embeddings, orange signifies textual embeddings without guidance, and green indicates textual embeddings with guidance, where (a) represents the results of the model $a7$, and (b) represents the results of the model full.

model using only global feature significantly outperforms the one using only salient feature. This is because global feature generally encompasses all information in the RS images, while salient feature filters out some information, leading to an overall decline in retrieval performance of the model. The comparison between $a1$ and $a7$ results underscores the efficacy of global feature; at the same time, the comparison between $a3$ and $a7$ results indicates the effectiveness of salient features. The comparison between $a5$ and the full results highlights the effectiveness of the VGSM. The contrast of $a6$ and the $full$ results demonstrates the efficacy of the TGEM. Similarly, juxtaposing $a7$ and the full results reveals the potency of the MVGF module.

E. Visualization of Feature

In order to provide a more intuitive reflection of the impact that the proposed MVGF module has on cross-modal semantic alignment, we reduce the dimensionality of image and text features, mapping them onto a 2-D plane for observation. Specifically, we employ the t-distributed stochastic neighbor embedding (t-SNE) algorithm [55], a machine learning algorithm for dimensionality reduction. It endeavors to align the similarity of points in the high-dimensional space with their equivalents in the low-dimensional space, thereby preserving the local structure of data while concurrently uncovering significant global structures.

We conducted experiments across the entire RSITMD test set, selecting models $a7$ and full from the ablation study for a comparative visualization of features. Model full represents the complete VGSGN architecture, while model $a7$ lacks the MVGF module compared to full; that is, model $a7$ does not utilize visual global and salient features to guide text features. The visualization of features for $a7$ is shown in Fig. 10(a), whereas the full model, which employs the guidance of visual global and salient features for text features, is depicted in Fig. 10(b). Compared to the results without visual feature guidance, the inclusion of the MVGF module significantly narrows the heterogeneous gap between visual and text features, facilitating cross-modal feature alignment. The visualization results intuitively demonstrate the efficacy of visual guidance in fostering cross-modal feature alignment between image and text.

TABLE III
QUANTITATIVE RESULTS OF INTRA-CLASS CENTER DISTANCE

Model	Intra-class center distance		
	Min distance	Max distance	Mean distance
$a7$	3.61	8.25	5.61
full	0.52	1.78	1.08

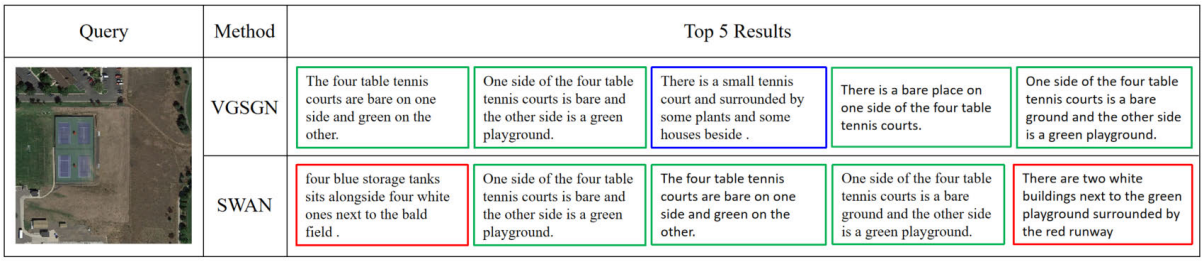
Furthermore, in order to objectively evaluate the impact of the MVGF module on cross-modal feature alignment, we further introduce more specific category labels based on the two major categories of image and text. For the RSITMD test set used, classification is based on naming, and images and texts include 32 corresponding categories, respectively. The experiments are conducted in the initial high-dimensional feature space. For each category, use k-means clustering algorithm to obtain the image clustering center and text clustering center, respectively, then calculate the intra-class center distance, and calculate the minimum, maximum, and average value of the intra-class center distance. The results are as shown in Table III. It can be seen from the results that, after the introduction of the MVGF module, the intra-class center distance is shortened by 80.7% on average, contributing to achieving improved image–text matching results.

F. Result Analysis of Retrieval Tasks

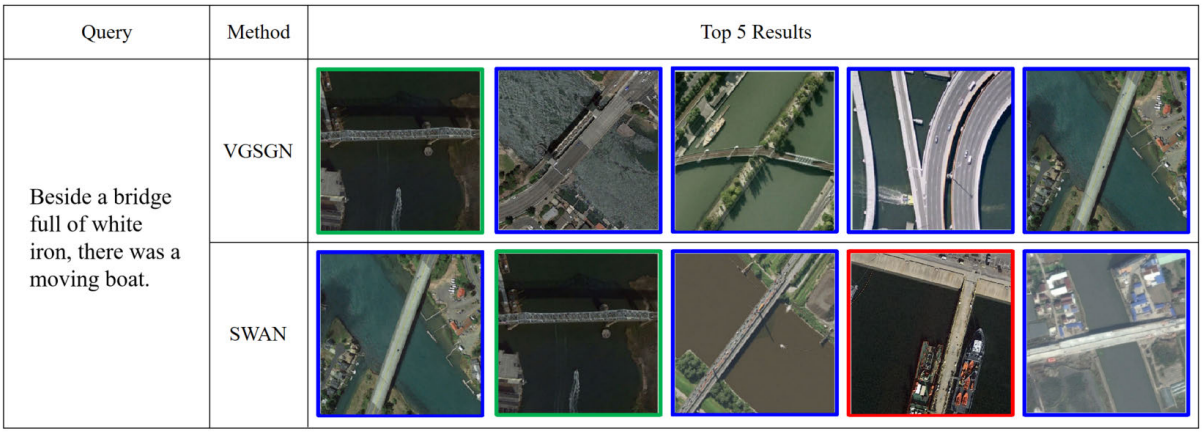
In this section, we present visual analysis of typical retrieval results to intuitively analyze the performance differences among different models. We chose the previously top-performing SWAN model as a reference for comparison with our method. SWAN has introduced a fine-grained perception module to enhance scene awareness and reduce semantic confusion. We utilize these two approaches for the tasks of image-to-text retrieval and text-to-image retrieval on the RSITMD test set, as depicted in Fig. 11.

The results of the image-to-text retrieval task are illustrated in Fig. 11(a). The VGSGN method contains four true values among the five retrieval results. Although the results that ranked three do not match, the key entity it describes is also of the same class as the query image. In contrast, the five retrieval results of the SWAN method only contain three true values, while the key entities described by the results of rankings 1 and 5 are not of the same class as the query image. The results of the text-to-image retrieval task are shown in Fig. 11(b). The five search results of the VGSGN method are all of the same class as the key entities described in the query text, and the result ranked 1 is the true value. Although the result of SWAN ranking 2 is true value, the result of ranking 4 is not of the same category as the key entity described in the query text. The results of the two retrieval tasks indicate that our proposed VGSGN has stronger semantic alignment ability and can achieve better image–text matching results.

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(a)



(b)

Fig. 11. Retrieval results of VGSGN and SWAN on the RSITMD test set. We display the top five retrieved results. The green box indicates the correct matching result, the blue box represents the result that is not correctly matched but belong to the same category, and the red box indicates the result that is neither correctly matched nor belong to the same category. (a) Results for the image-to-text retrieval task. (b) Results for the text-to-image retrieval task.

G. Comparative Results of Semantic Localization

In this section, to exhibit the practical application capability of VGSGN, we conducted semantic localization experiments. The semantic localization task refers to the process of pinpointing the scene that best matches the input query text within a large RS scene. Following the approach in [56], a multiscale sliding window is employed to segment the large scene RS image, then compute the similarity between the query text and each segment, and integrate these similarities at the pixel level. Finally, utilizing a median filter to eliminate impulse noise in the probability map, yielding the more robust results.

We qualitatively compare the visualization results of VGSGN with those of the leading competitor SWAN method and the classical AMFMN method. The visual results are depicted in Fig. 12, where the first row comprises the visualized outcomes of semantic localization, while the second row displays the heat maps, with red indicating higher likelihood and blue indicating the opposite. We train three models on the RSITMD dataset and conduct retrieval using the input query "a bridge spans a dark blue river" on large scene RS image. In Fig. 12 (from left to right), we present the results of AMFMN (a), SWAN (b), and VGSGN (c), respectively. By integrating the three sets of localized result images with the corresponding heat maps, it becomes evident that all three models successfully retrieved the queried subject. Our proposed VGSGN model demonstrates the most focused results, followed by SWAN, with AMFMN performing the least satisfactorily. This observation signifies the superior semantic alignment capability of VGSGN.

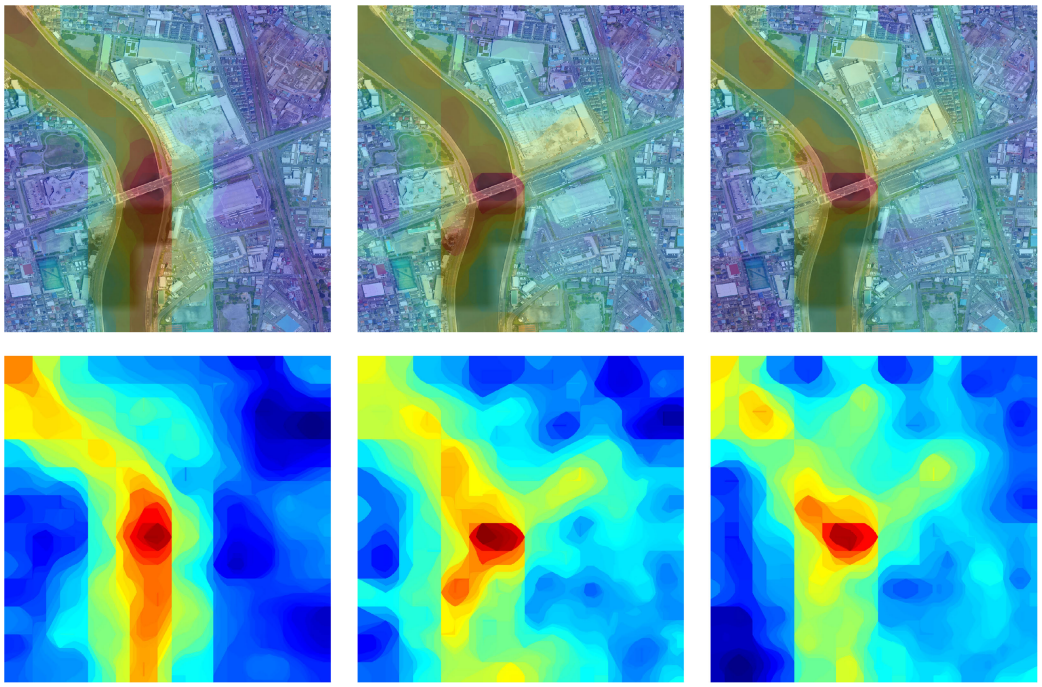
H. Comparisons With CLIP-Based Methods

In this section, we compare the mean recall of our method with CLIP-based methods (out domain defined in [36]), as indicated in Table IV. The mean recall, defined as the average of R@1, R@5, and R@10, is used to measure the overall performance for both sentence retrieval and image retrieval. CLIP-original denotes the original CLIP model. CLIP-laion-RS denotes training a CLIP model on the LAION-RS dataset (a subset of LAION-2B [57]). SkyCLIP denotes training a CLIP model on the SkyScript dataset. The results show that the retrieval performance of our method outperforms the CLIP-based methods (out domain), which also indicates the effectiveness of our problem-solving ideas. The captions in the RSICD and RSITMD datasets are short and just providing simple descriptions of the corresponding RS images. The use of large language models for these descriptions could not extract significantly more meaningful textual information. In addition, the size of the datasets used is not suitable for fine-tuning the large models and the compatility consumption associated with the use of large models. The retrieval performance of the CLIP-based methods on the RSICD and RSITMD datasets has no significant advantage.

I. Analysis of Time Consumption

In this section, we compare the retrieval time consumption of our method with other excellent methods. The experimental platform includes an Intel Core i9-12900KF CPU @3.10 GHz and a single NVIDIA RTX 3090 GPU. We conduct multiple experiments with no other load on the equipment and take the average as the final result to ensure the reliability of the

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(a) (b) (c)

Fig. 12. Visualization comparison of semantic localization with retrieval of "a bridge spans a dark blue river." (a) Semantic localization outcomes of AMFMN. (b) Semantic localization outcomes of SWAN. (c) Semantic localization outcomes of VGSGN.

TABLE IV
COMPARISON OF MEAN RECALL WITH CLIP-BASED METHODS

Model	RSICD		RSITMD	
	img2txt	txt2img	img2txt	txt2img
CLIP-original(ViT-32)	19.67	13.84	27.51	24.10
CLIP-original(RN-50)	20.61	19.54	31.36	29.97
CLIP-laion-RS	22.66	18.52	30.24	29.67
SkyCLIP	**23.70**	19.97	30.75	30.58
Ours	20.90	**22.17**	**33.26**	**39.58**

TABLE V
TIME CONSUMPTION OF DIFFERENT METHODS

Approach	Evaluation time(s)		Params(M)	FLOPs(G)
	RSICD	RSITMD		
AMFMN	45.75	11.10	36.70	3.41
GaLR	47.43	11.79	46.89	3.42
SWAN	62.58	13.05	40.46	6.63
CLIP-original(RN-50)	1153.64	278.72	149.62	69.24
VGSGN	64.16	13.58	54.31	6.75

results. We conduct multiple experiments on the experimental platform and take the average as the final result to ensure the reliability of the results. The comparison results are shown in Table V.

As shown in Table V, VGSGN exhibits a slightly higher time consumption compared to GaLR and AMFMN, on par with SWAN. VGSGN showcases 16% enhancement in accuracy relative to GaLR and 7% improvement compared to SWAN on the RSITMD test set. By comparing the above methods, our method attains superior performance with an acceptable increase of retrieval time consumption, network's parameters, and floating point operations (FLOPs). Meanwhile, our method demonstrates obvious advantages in terms of computation complexity and evaluation time expenditure compared to CLIP-original (RN-50). The CLIP-based methods have a similarly evaluation time and FLOPs, our method should have the same advantages compared to them.

V. CONCLUSION

In this article, we address the challenge of feature alignment due to the heterogeneity between image and text modal data. We contemplate structurally sufficiently utilizing the image feature to guide text feature. Hence, we introduce VGSGN. VGSGN achieves improved retrieval performance by focusing on perceiving global and salient information in RS images, redundant filtering and updating enhancement of text features, and leveraging image-guided text to help align image and text features. In visual representation, we design VGSM to acquire deep visual global feature and shallow visual salient feature. Then dynamically fusing global features with salient features yields a visual aggregated representation. In textual representation, we propose TGEM to filter redundant text information and update and enhance text feature. Subsequently, we design the MVGF module, using visual global and salient features to guide text feature and dynamically fuse them to obtain guided and enhanced text feature. This enhancement bolsters the semantic coherence between image and text modal features, aiding in feature alignment. The quantitative analysis results on multiple RS text–image datasets, along with the findings from ablation experiments and retrieval tasks, confirm the effectiveness and superiority of VGSGN for RS retrieval.

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In addition, we also compare our method with the CLIP-based methods, and the results show that our method outperforms the CLIP-based methods (out domain) in retrieval performance. If fine-tuning of large models is considered, this is another research direction. Currently fine-tuning large model has become a research hotpot. The quality of the fine-tuning method will have a direct impact on the performance of the fine-tuned large model. Achieving highly effective fine-tuning is fraught with challenges and requires in-depth research. The computational complexity of the large model-based method in the evaluation process is higher than our method, which is a factor worthy of consideration in practical applications. The information interaction approach proposed in this article may provide new ideas for the current framework of large models.

Currently, commonly used RS image–text datasets such as RSICD and RSITMD have relatively simple textual descriptions that cannot provide sufficient text information for corresponding RS images. In future research work, to improve retrieval performance, it may be deemed necessary to establish a dataset with comprehensive textual descriptions.

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